

Track 2 Official Model Verification Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Verdict

This update accepts the completed `Track 2H` robust-loss dispersion-aware campaign into the official `Track 2` offline verification matrix.

Decision

- `Track 2H` is verified as an exploratory robust-loss baseline.
- No `Track 2H` candidate is promoted over the current accepted `Track 2` leaders.
- The project continues to maintain three parallel best surfaces: `Fw` , `Bw` , and `global` .
- Within `Track 2H` , the strongest forward candidate is `track2h_mae_robust_Fw` .
- Within `Track 2H` , the strongest backward candidate is `track2h_smooth_l1_robust_Bw` .
- Within `Track 2H` , the strongest global candidate is `track2h_mae_robust_global` .

Source Package

This official report consolidates these refreshed artifacts:

- metric matrix:
doc/reports/analysis/track2/Track 2 Directional Model Comparison.md ;
- matrix summary:
output/validation_checks/track2_reference_comparison/2026-06-11-15-04-04__track2_full_directional_family_matrix_track2h_robust_loss_track2_refresh_2026_06_11/validation_summary.yaml ;
- per-condition metrics:
output/validation_checks/track2_reference_comparison/2026-06-11-15-04-04__track2_full_directional_family_matrix_track2h_robust_loss_track2_refresh_2026_06_11/per_condition_metrics.csv ;
- validation report:
doc/reports/analysis/validation_checks/track2/2026-06-11-15-14-38_track2_full_directional_family_matrix_track2h_robust_loss_track2_refresh_2026_06_11_report.md ;
- best-model collage report:
doc/reports/analysis/track2/best_model_collage_report/[2026-06-11]/track2_best_model_collage_report.md ;
- multi-model curve comparison report:
doc/reports/analysis/track2/multi_model_curve_comparison_report/[2026-06-11]/track2_multi_model_curve_comparison_report.md ;
- operator launch logs:
output/validation_checks/track2_operator_launch_logs/2026-06-11-15-04-00_track2h_robust_loss_track2_refresh_2026_06_11/ .

Candidate Refresh

The refresh added 9 registry-backed Track 2H candidates covering robust MAE , SmoothL1 , and LogCosh losses across the global , Fw , and Bw surfaces.

Surface	Robust Loss	Candidate
global	MAE	track2h_mae_robust_global
Fw	MAE	track2h_mae_robust_Fw
Bw	MAE	track2h_mae_robust_Bw
global	SmoothL1	track2h_smooth_l1_robust_global
Fw	SmoothL1	track2h_smooth_l1_robust_Fw
Bw	SmoothL1	track2h_smooth_l1_robust_Bw
global	LogCosh	track2h_log_cosh_robust_global
Fw	LogCosh	track2h_log_cosh_robust_Fw
Bw	LogCosh	track2h_log_cosh_robust_Bw

The matrix source group is track2h_dispersion_aware_modeling_registry . The matrix now contains 141 candidates.

Verification Rules

Surface	Training or Archive Scope	Evaluation Scope
Fw	forward-only training or archive	forward Track 2 curves only
Bw	backward-only training or archive	backward Track 2 curves only
global	forward and backward training together	both directions, reported by direction and combined

The `global` , `Fw` , and `Bw` branches are carried forward in parallel. They are not collapsed into one scalar winner.

Current Leaders

Overall Direction Leaders

Direction	Current Strongest Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
forward	<code>rcim_retuned_GBM19_Fw</code>	0.001089	0.001299	2.372	4.912
backward	<code>periodic_gru_sequence_Bw</code>	0.002392	0.002639	5.466	14.820
global combined	<code>periodic_gru_sequence_global</code>	0.002704	0.002949	6.139	13.200

Repository-Owned Static Baselines

Direction	Current Strongest Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
backward	<code>tree_Bw</code>	0.003258	0.003651	7.051	14.116
global combined	<code>tree_global</code>	0.003144	0.003533	6.854	13.314

Repository-Owned Neural Leaders

Surface	Current Strongest Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
Bw	<code>periodic_gru_sequence_Bw</code>	0.002392	0.002639	5.466	14.820
global combined	<code>periodic_gru_sequence_global</code>	0.002704	0.002949	6.139	13.200

Track 2H Result

Surface	Candidate	Robust Loss	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
Bw	<code>track2h_smooth_l1_robust_Bw</code>	<code>SmoothL1</code>	0.003078	0.003403	6.676	15.410
Fw	<code>track2h_mae_robust_Fw</code>	<code>MAE</code>	0.003134	0.003382	6.956	12.470
Fw	<code>track2h_smooth_l1_robust_Fw</code>	<code>SmoothL1</code>	0.003300	0.003545	7.342	13.401

Surface	Candidate	Robust Loss	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
Fw	track2h_log_cosh_robust_Fw	LogCosh	0.003344	0.003595	7.427	12.598
global combined	track2h_mae_robust_global	MAE	0.003401	0.003715	7.504	13.873
global combined	track2h_smooth_l1_robust_global	SmoothL1	0.003417	0.003719	7.539	14.332
Bw	track2h_mae_robust_Bw	MAE	0.003433	0.003750	7.506	14.575
Bw	track2h_log_cosh_robust_Bw	LogCosh	0.003486	0.003811	7.628	12.872
global combined	track2h_log_cosh_robust_global	LogCosh	0.003498	0.003819	7.697	14.405

Robust losses improve the previous Track 2G branch on the best Fw and Bw surfaces. The strongest Fw Track 2H candidate improves over the best Track 2G Fw candidate from 0.003174 to 0.003134 MAE. The strongest Bw Track 2H candidate improves over the best Track 2G Bw candidate from 0.003436 to 0.003078 MAE. The best Track 2H global candidate remains slightly behind the best Track 2G global candidate.

Visual Evidence

The 2026-06-11 collage and overlay bundles were regenerated with dedicated Track 2H sections:

- Forward Track 2H Robust-Loss Models ;
- Backward Track 2H Robust-Loss Models ;
- Global Track 2H Robust-Loss Models in the collage report;
- Forward Reference Tree And Track 2H Overlay ;
- Backward Reference Tree And Track 2H Overlay .

The visual package supports the matrix decision: robust losses are useful controls for dispersion-aware training, especially in the backward direction, but they do not yet replace the strongest periodic temporal branch.

Campaign Update Ledger

Date	Campaign or Update	Candidate Scope	Matrix Status	Visual Status	Decision
2026-06-11	Track 2H robust-loss dispersion-aware refresh	9 robust global , Fw , and Bw candidates	included in the 141 - candidate matrix	dated collage and overlay bundles regenerated with Track 2H sections	verified exploratory baseline; not promoted
2026-06-10	Track 2G curve-aware training refresh	12 pointwise-control, centered-shape, offset, and full-composite global , Fw , and Bw candidates	included in the 132 - candidate matrix	dated collage and overlay bundles regenerated with Track 2G sections	verified exploratory baseline; not promoted
2026-06-08	Track 2F-bis harmonic-offset probe refresh	6 clean and harmonic global , Fw , and Bw candidates plus 3 rechecked Track 2F candidates	included in the 120 - candidate matrix	dated collage and overlay bundles regenerated with Track 2F-bis sections	verified exploratory baseline; not promoted

Date	Campaign or Update	Candidate Scope	Matrix Status	Visual Status	Decision
2026-06-04	Track 2F offset-aware probe refresh	3 global , Fw , and Bw sequential residual offset candidates	included in the 114 - candidate matrix	dated collage and overlay bundles regenerated	verified exploratory baseline; not promoted
2026-05-28	Wave 2C residual harmonic temporal hybrid refresh	18 global , Fw , and Bw residual harmonic GRU/LSTM candidates	included in the 111 - candidate matrix	collage and overlay reports refreshed with Wave 2C sections	verified exploratory baseline; not promoted over Wave 2B or accepted Track 2 baselines
2026-05-26	Wave 2B harmonic temporal hybrid refresh	periodic temporal convolution, GRU, and LSTM global , Fw , and Bw candidates	included	collage and overlay reports refreshed	strongest repository-owned neural branch

Closeout Decision

Track 2H does not change the accepted Track 2 baseline. The current direction-parallel decision remains:

- Fw : rcim_retuned_GBM19_Fw remains the strongest overall forward candidate; track2h_mae_robust_Fw is the strongest Track 2H forward candidate and is slightly better than the best Track 2G forward candidate.
- Bw : periodic_gru_sequence_Bw remains the strongest practical repository-owned backward candidate; track2h_smooth_11_robust_Bw is the strongest Track 2H candidate and improves over the static tree_Bw baseline in Track 2 MAE.
- global : periodic_gru_sequence_global remains the strongest repository-owned bidirectional neural candidate; within Track 2H, the MAE robust global candidate is the strongest but remains behind the current Track 2G global leader and the accepted periodic temporal global branch. The next modeling step should keep robust losses as a validated ingredient, but not treat them as sufficient. The evidence supports preparing the next dispersion-aware package around quantile or probabilistic regression, followed by mixture-density heads and latent-state or hysteresis-aware variants before the integrated multi-task / multi-head model branch.